

DSC 411/511
Advance Data Mining Algorithms and Applications

Fall 2026

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Location: MSB 104



Lecture 09
Clustering
(Density Estimation & DBSCAN)

Today's class:

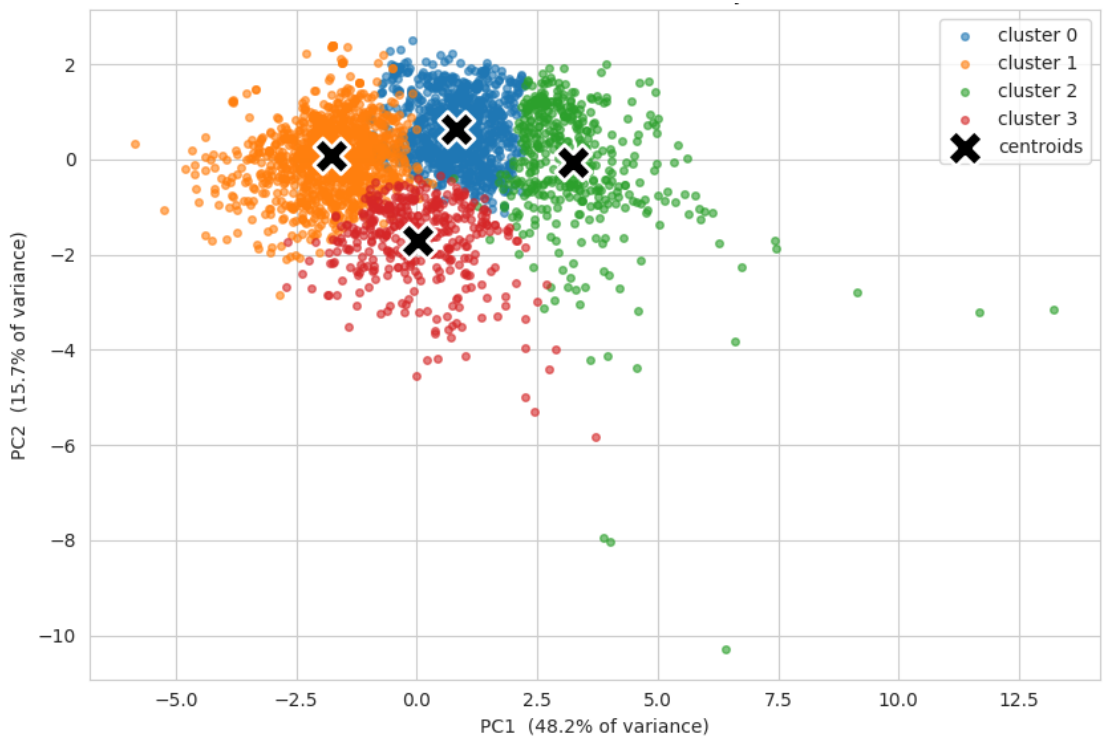
- ***Density Estimation***
- ***DBSCAN***

Density Estimation

Definition

Density estimation is the task of estimating, from a sample of points, how crowded each region of the space is.

Chapter 9
Introduction to Data Mining, 2nd Edition



- Where the crowd is?
- Where the gaps are?
- Point (6.5, −10.3) Assigned to cluster 2, why?

Density Estimation

Definition

Density estimation is the task of estimating, from a sample of points, how crowded each region of the space is.

Chapter 9
Introduction to Data Mining, 2nd Edition

In plain language

- Given a scatter of points, work out how many points are near any given spot.
- The output is not a group label. It is a number for every location.
- High number means crowded. Low number means empty.

It answers a different question?

- Classification asked: which category does this belong to?
- Clustering asked: which group does this belong to?
- Density asks: is this a crowded part of the space, or an empty one?

Density Estimation

Where it can be used?

- **Clustering:** A cluster is a dense region. The gaps between clusters are sparse.
- **Anomaly detection:** A point in a low-density region is unusual by definition.
- **Understanding one column:** Where does the data pile up, and where are the gaps?
- **Choosing a cut point:** Put the boundary in the valley, not at an arbitrary number.

Three ways to measure density

Method	How it works
Histogram	Divide into bins, count each one
Kernel Density Estimation	Put a bump on every point, add them up
Counting neighbours	Draw a circle of radius eps, count what is inside

Why this matters today

- k-means forced every house into a cluster.
- Hierarchical merged until nothing was left.
- Neither was allowed to say **"this point belongs to nothing."**

Density Estimation

Histograms

A histogram divides the range of a numeric attribute into intervals called bins, and reports how many values fall into each bin.

Section 3.3
Introduction to Data Mining, 2nd Edition



Density Estimation

Kernel Density Estimation

Kernel density estimation places a small function, the kernel, at every data point and sums them to produce a smooth estimate of the density across the whole range.

Chapter 9
Introduction to Data Mining, 2nd Edition

How it works, in three steps

- Put a small bump on every data point.
- Add all the bumps together.
- Where points are crowded, the bumps overlap and the total is high.

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

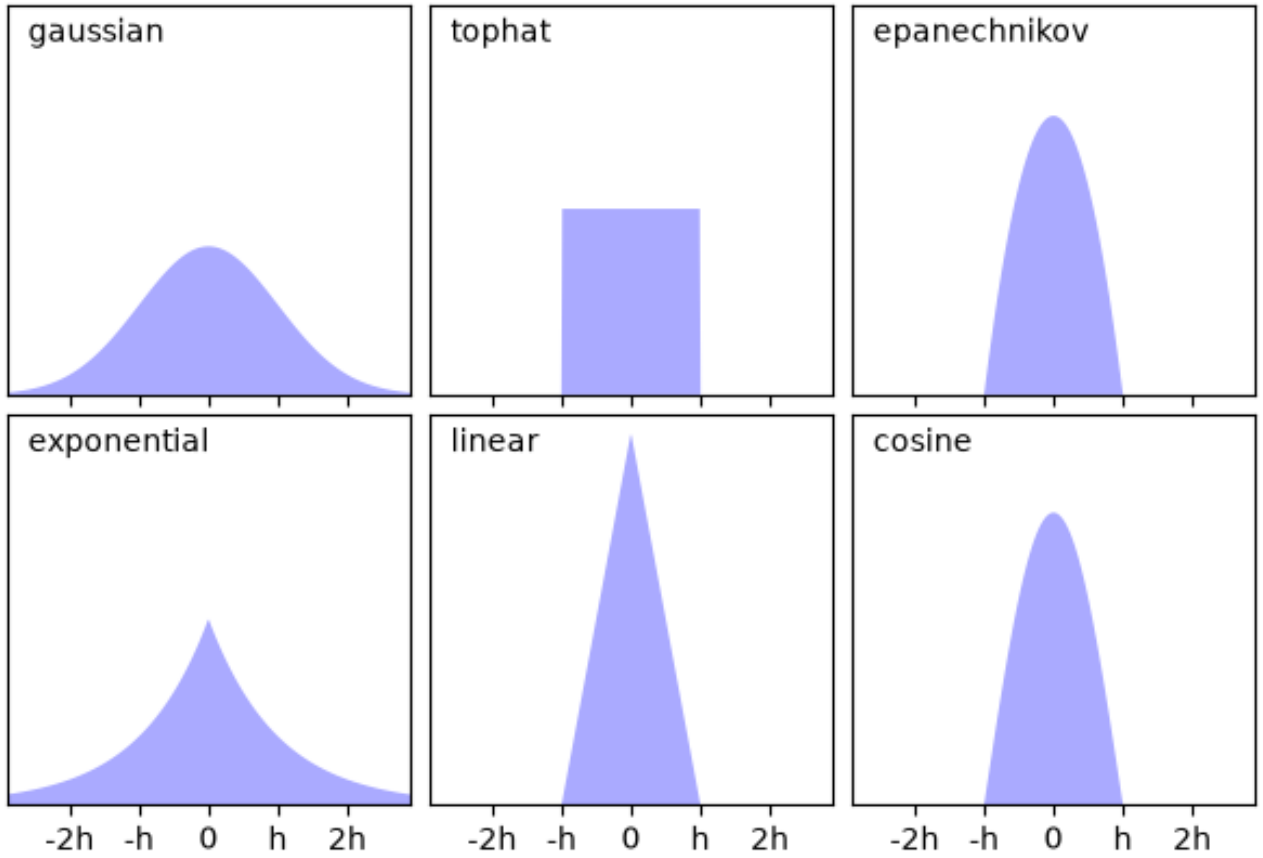
The one thing you choose: bandwidth (h)

- **h** is the width of each bump.
- Small h: every point becomes its own spike. Undersmoothed.
- Large h: everything merges into one hill. Oversmoothed.
- Nothing tells you the right h. It is your decision.

Density Estimation

How it works?

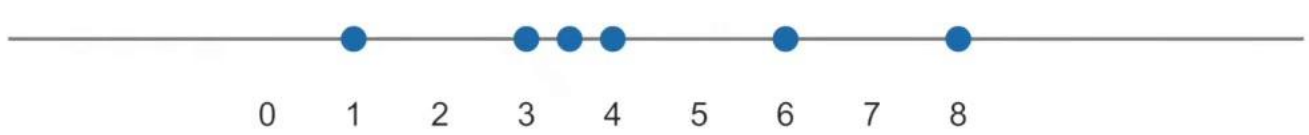
Different types of kernels



Density Estimation

How it works?

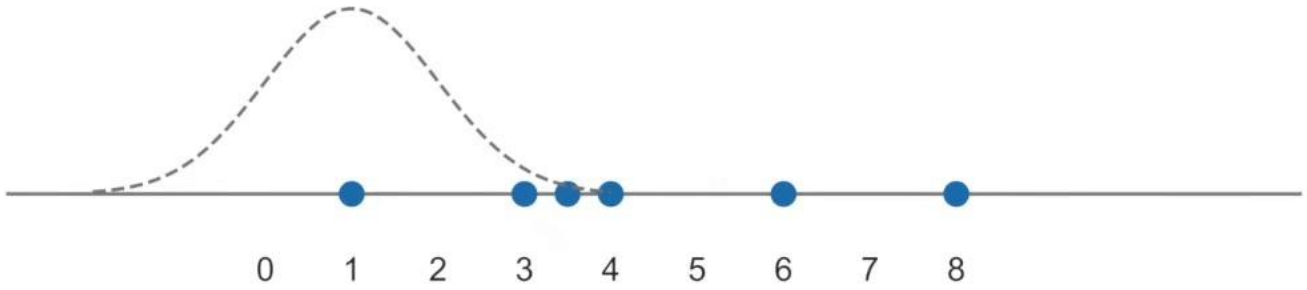
Step1: Put a small bump on every data point



Density Estimation

How it works?

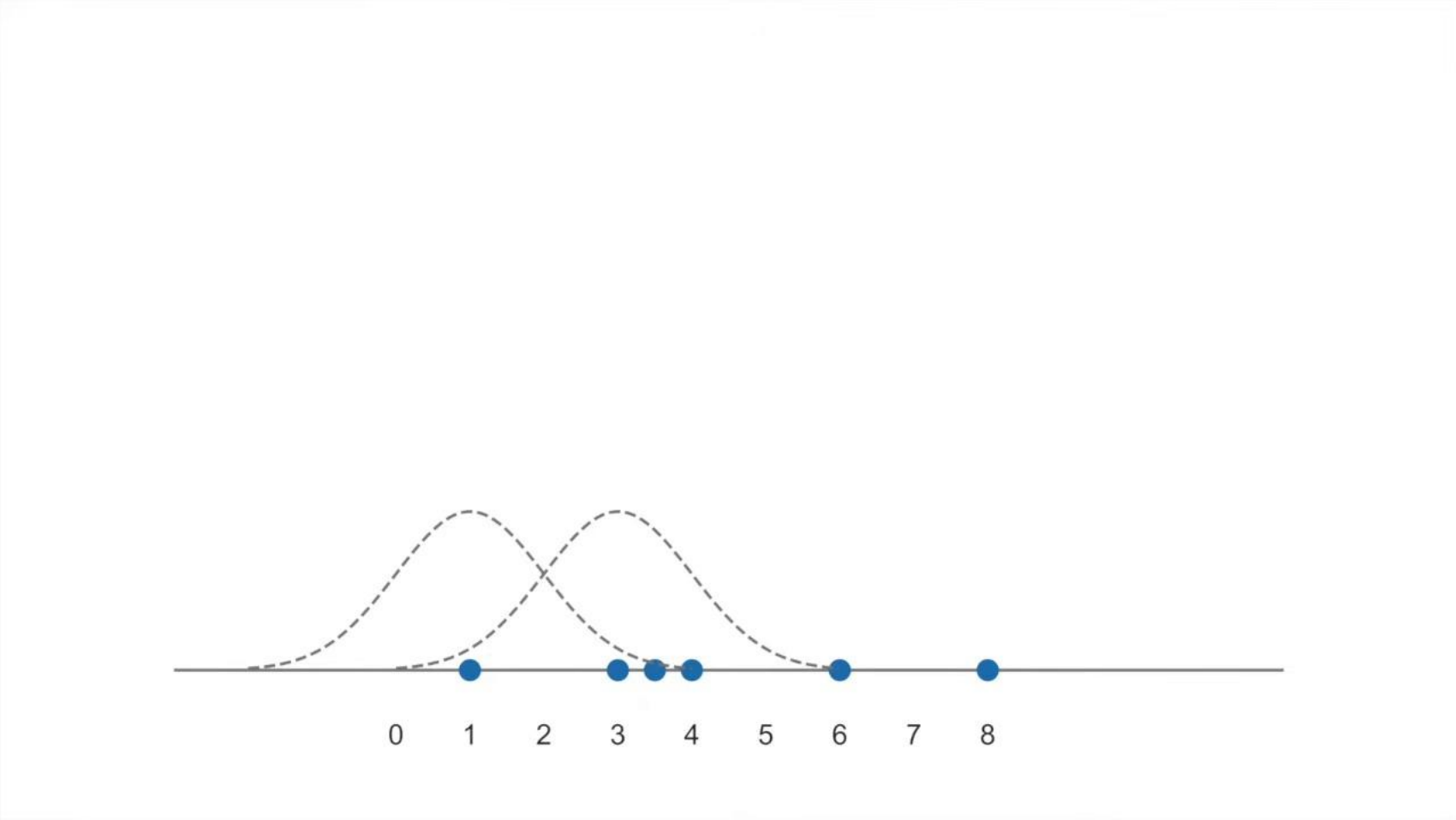
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Density Estimation

How it works?

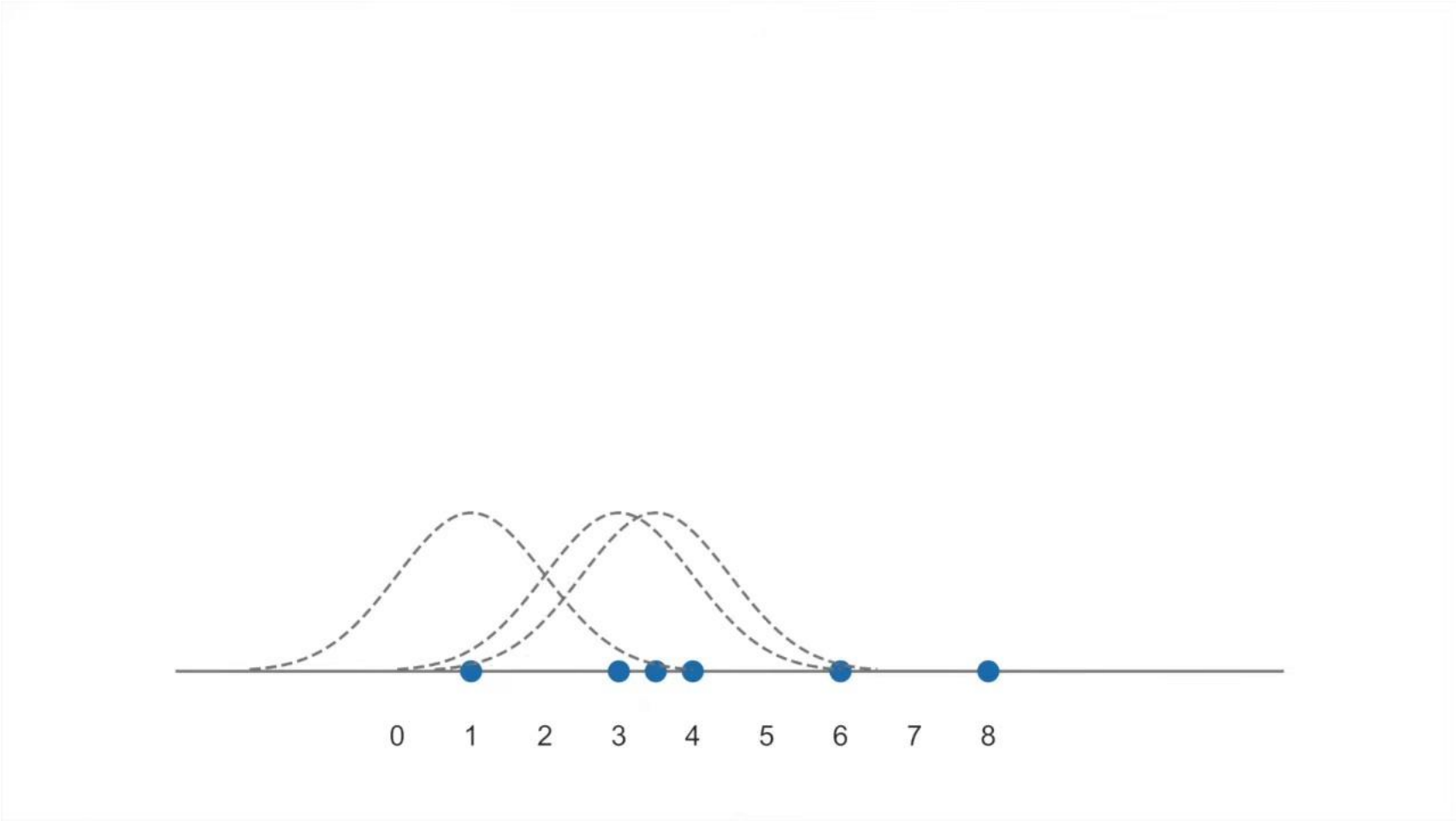
Step1: Put a small bump on every data point



Density Estimation

How it works?

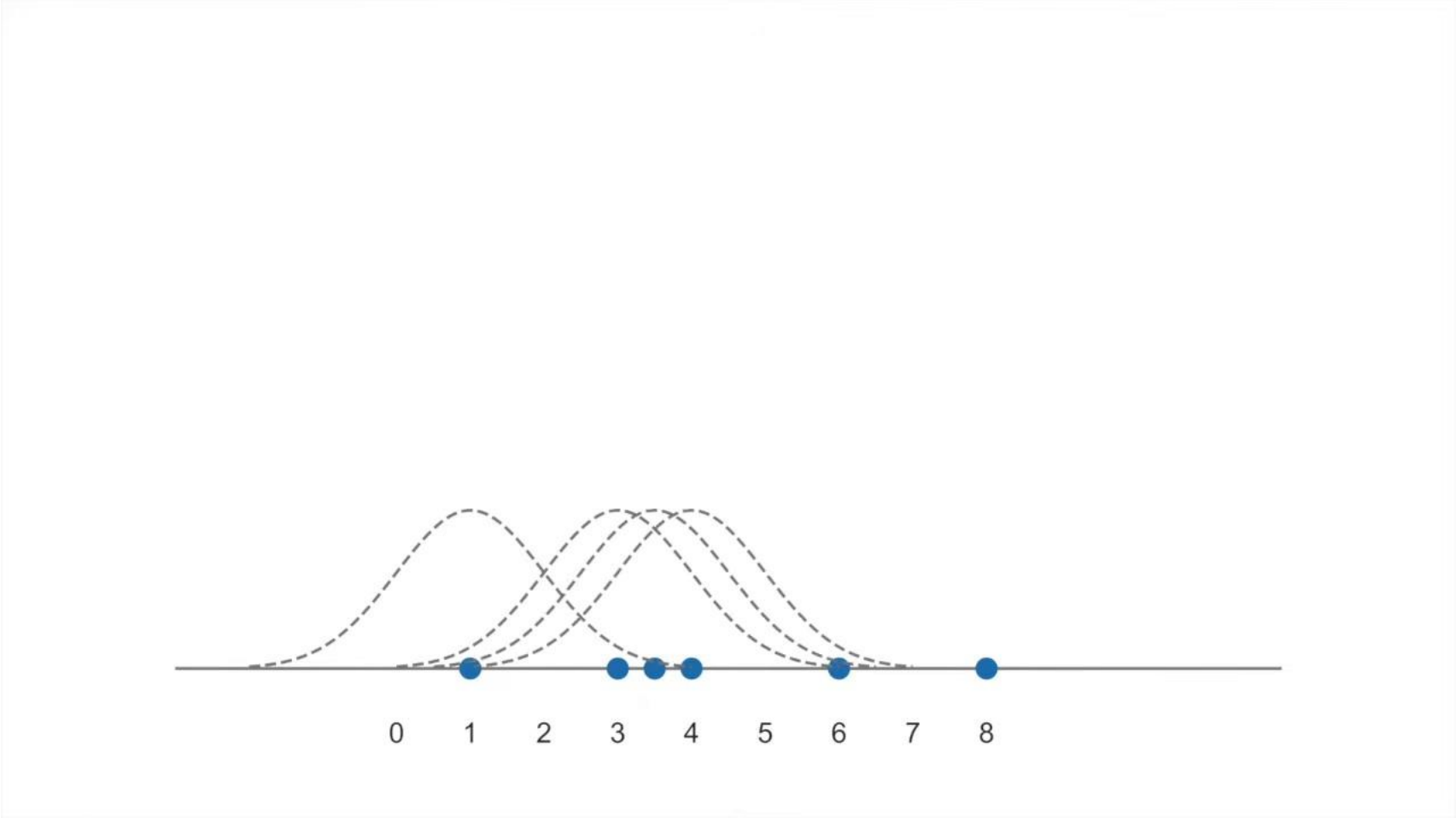
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Density Estimation

How it works?

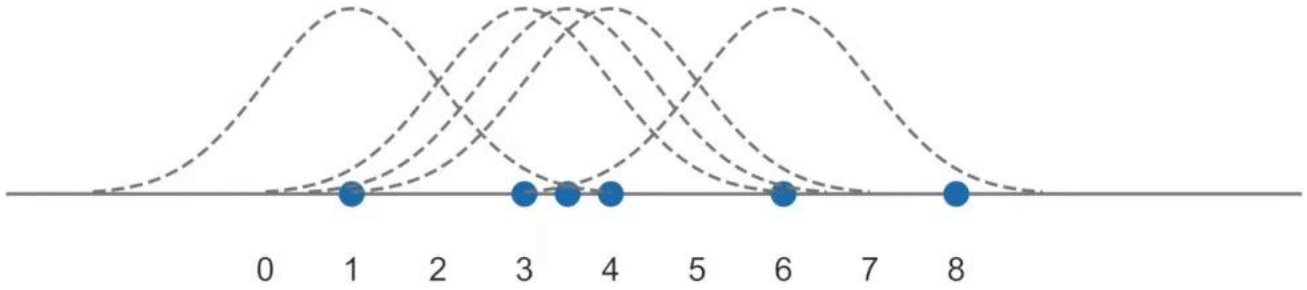
Step1: Put a small bump on every data point



Density Estimation

How it works?

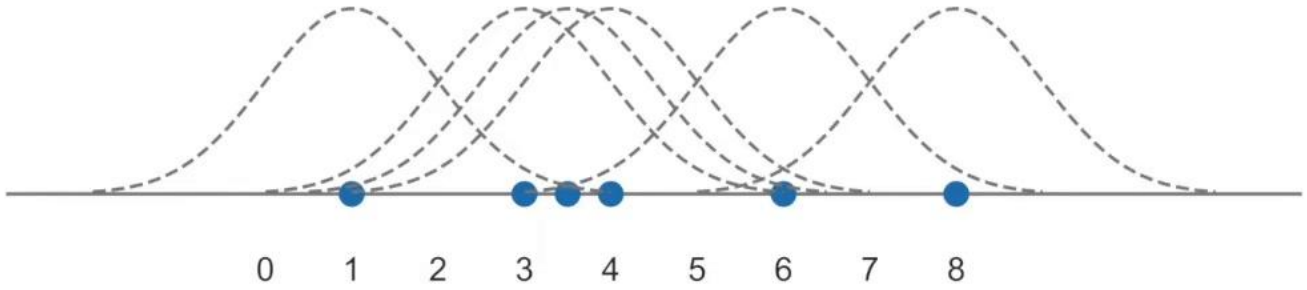
Step1: Put a small bump on every data point



Density Estimation

How it works?

Step1: Put a small bump on every data point

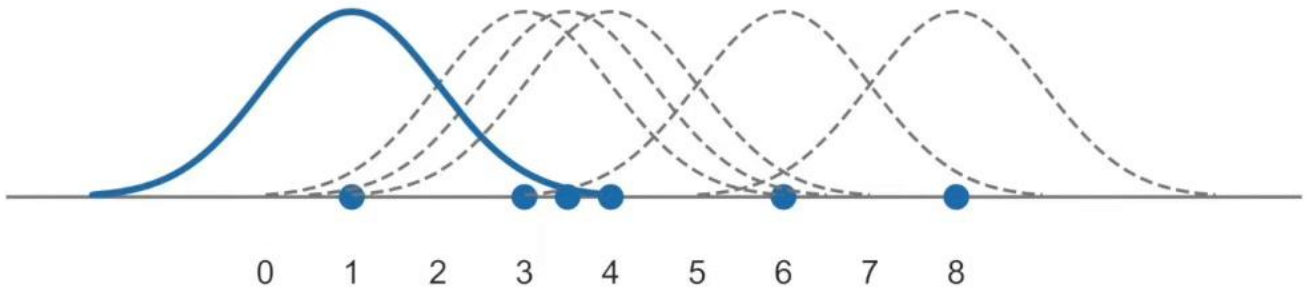


Density Estimation

How it works?

Step1: Put a small bump on every data point

Step2: Add all the bumps together.

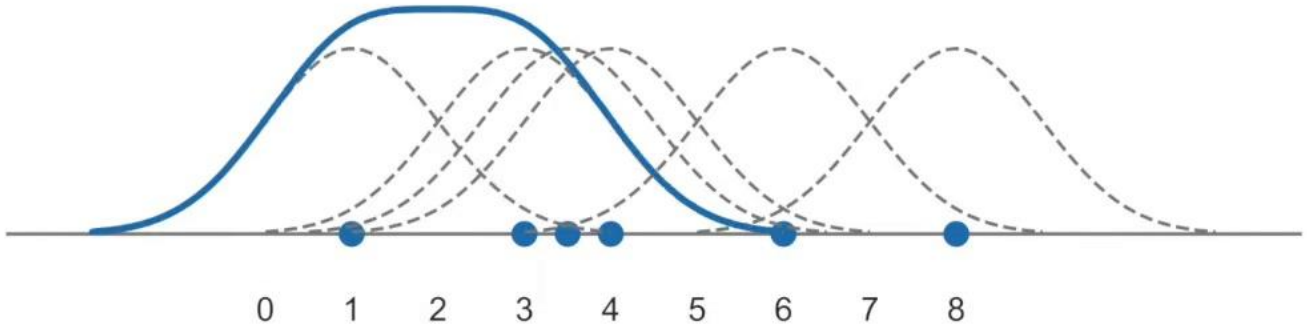


Density Estimation

How it works?

Step1: Put a small bump on every data point

Step2: Add all the bumps together.



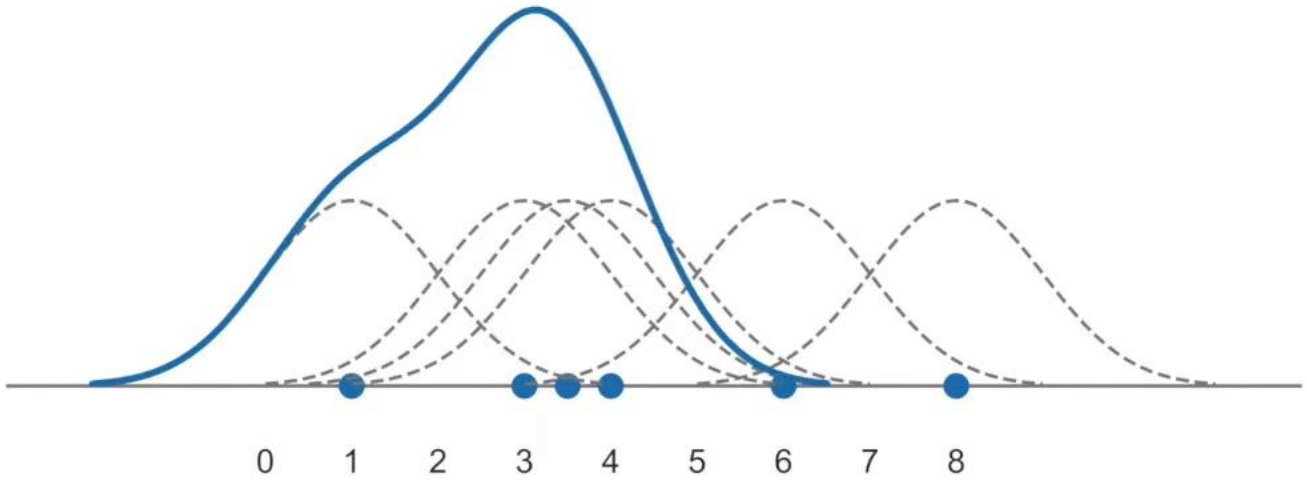
Density Estimation

How it works?

Step1: Put a small bump on every data point

Step2: Add all the bumps together.

Step3: Where points are crowded, the bumps overlap and the total is high.



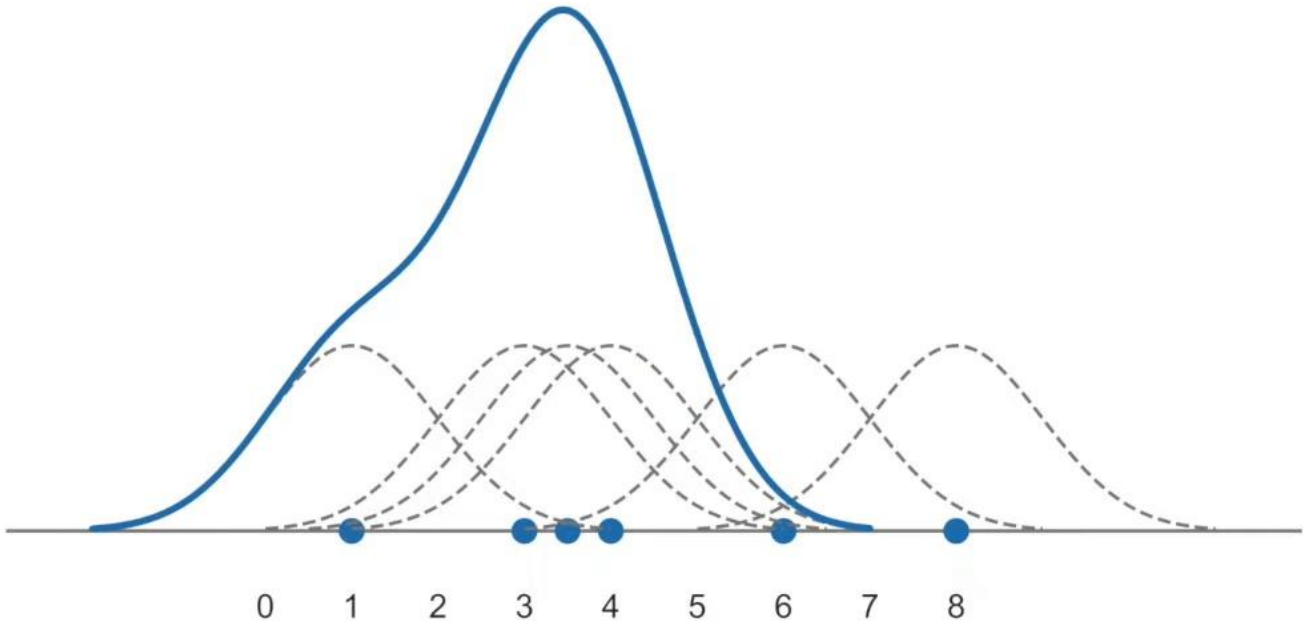
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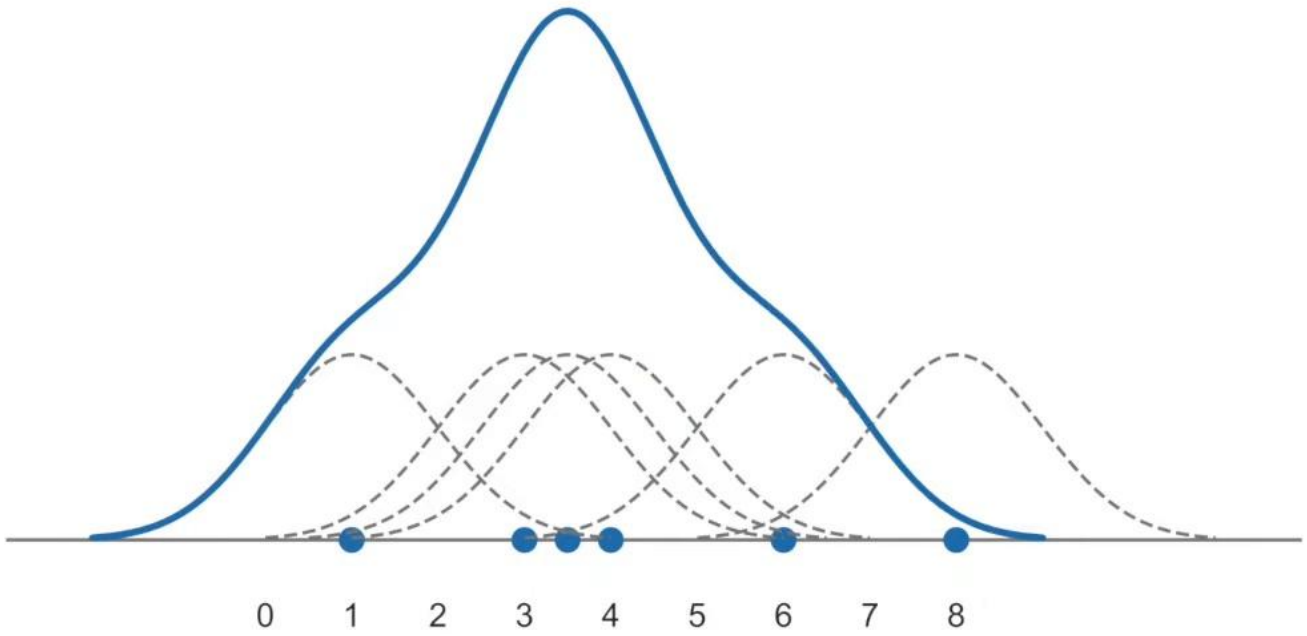
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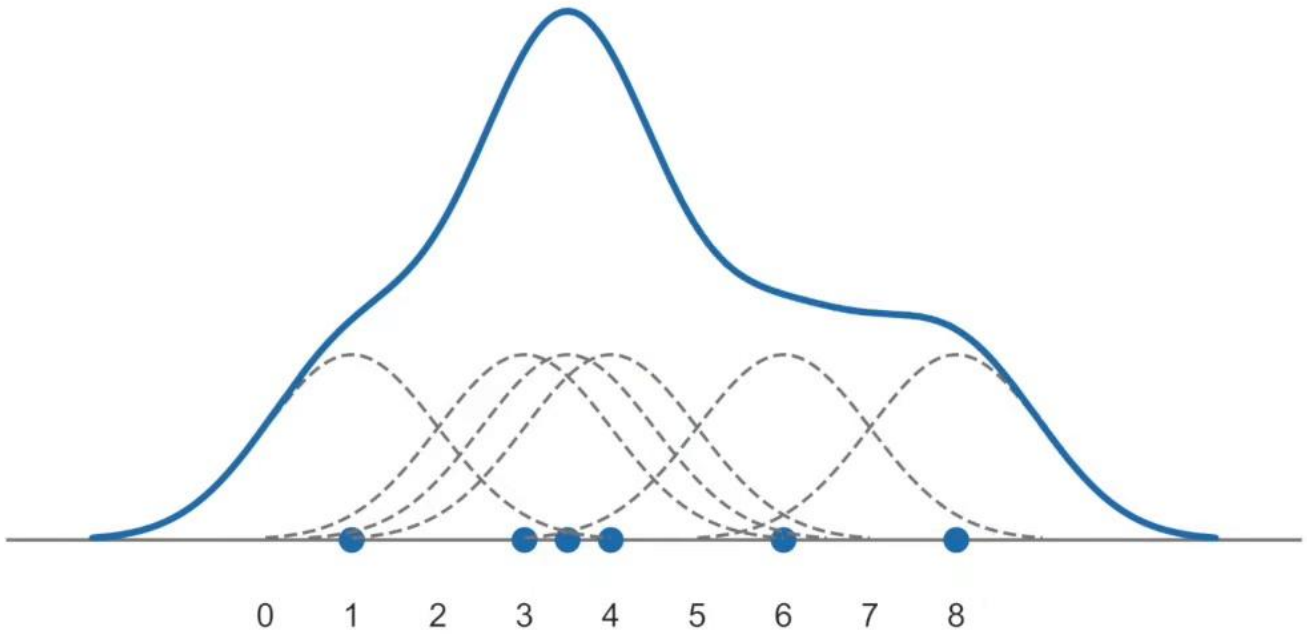
Density Estimation

How it works?

Step1: Put a small bump on every data point

Step2: Add all the bumps together.

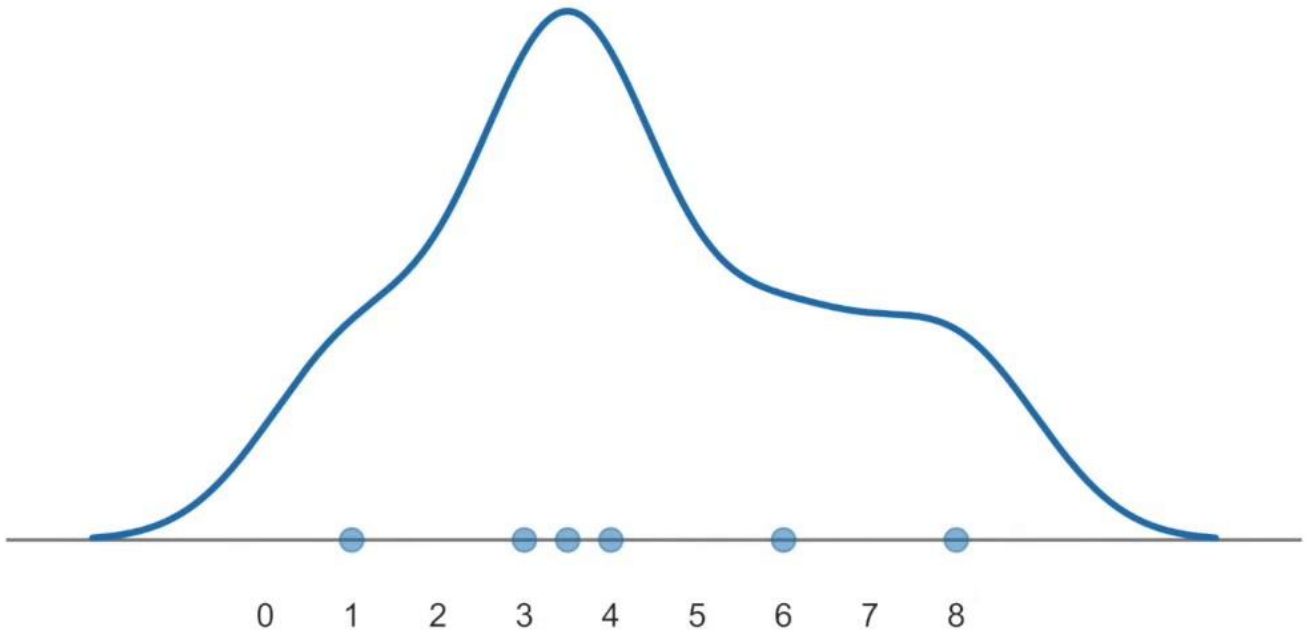
Step3: Where points are crowded, the bumps overlap and the total is high.



Density Estimation

How it works?

This is the density distribution of these data points.

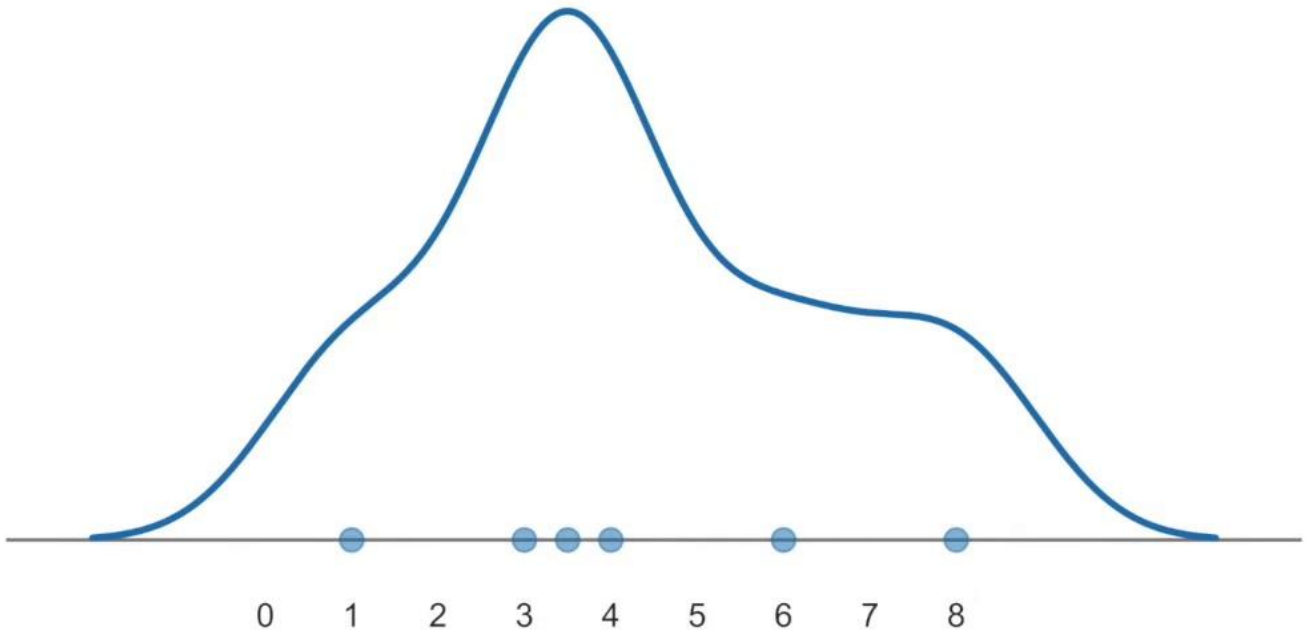


Density Estimation

How it works?

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

h=1.0

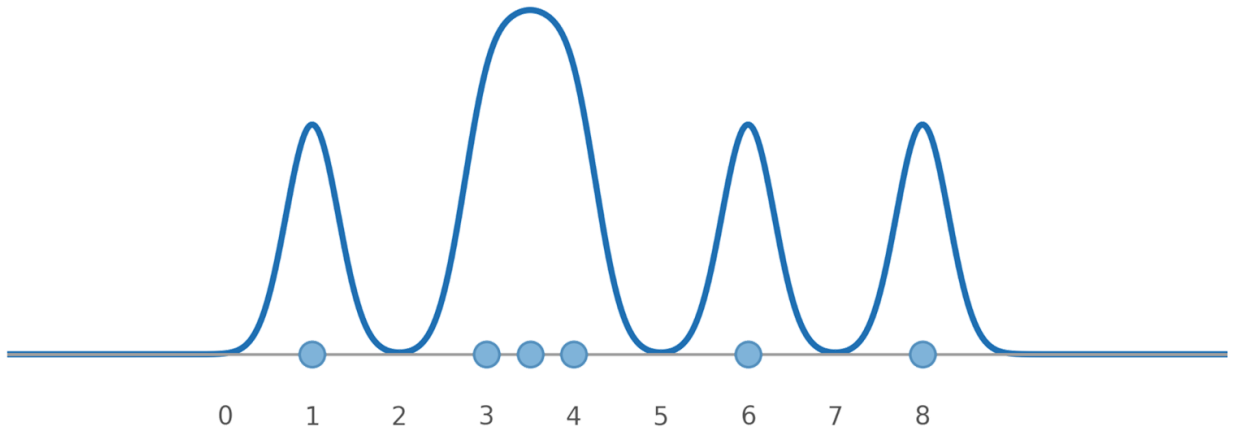


Density Estimation

How it works?

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

h=0.3

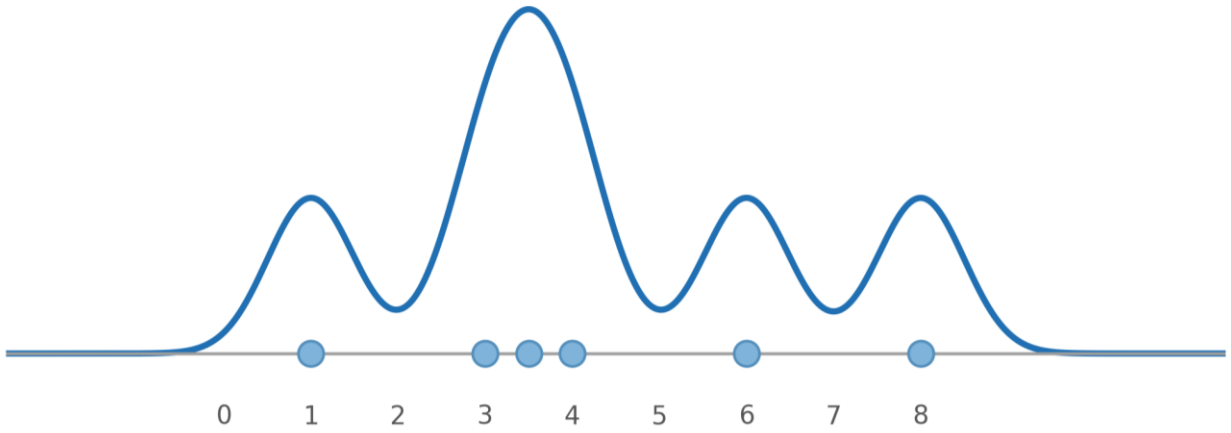


Density Estimation

How it works?

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

h=0.5

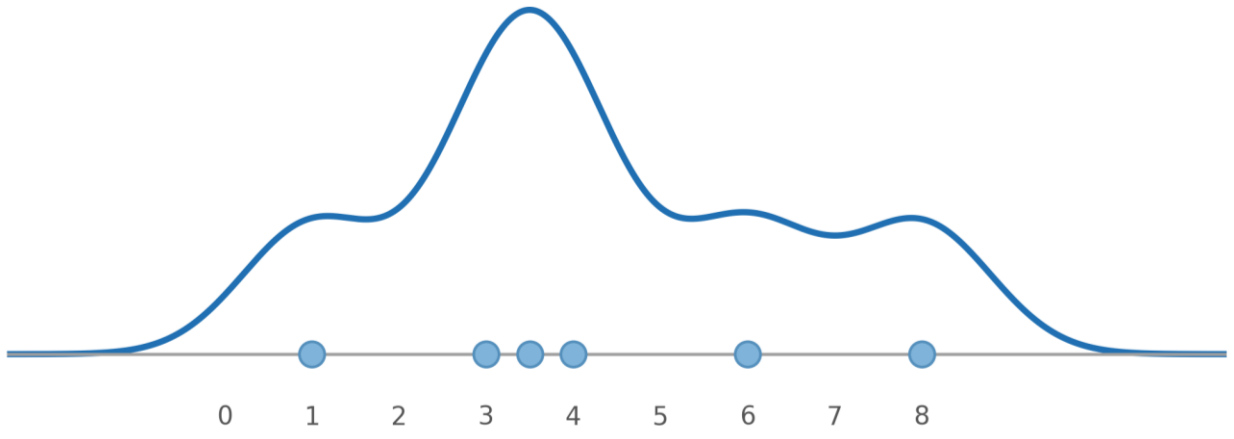


Density Estimation

How it works?

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

h=0.8

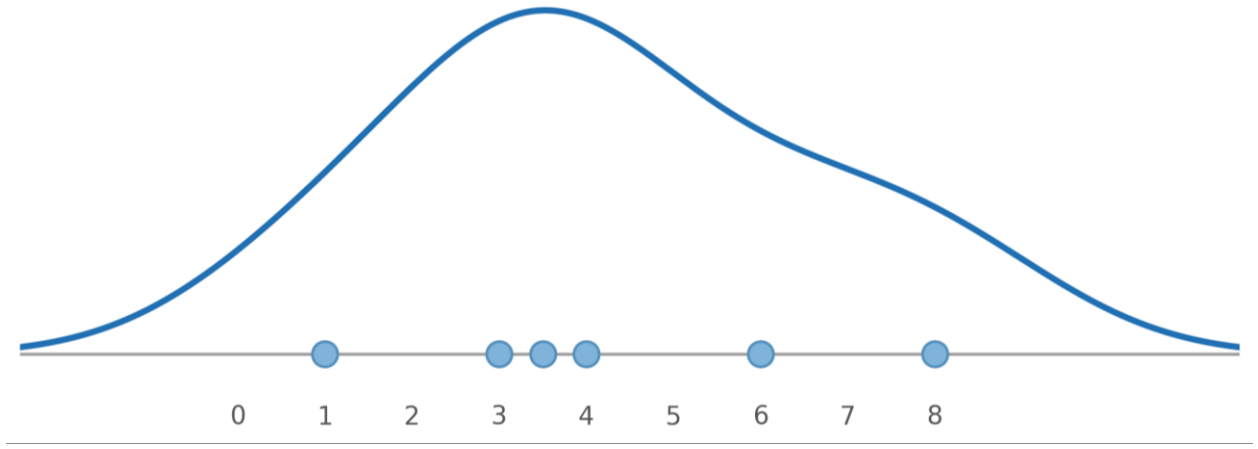


Density Estimation

How it works?

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

h=1.5

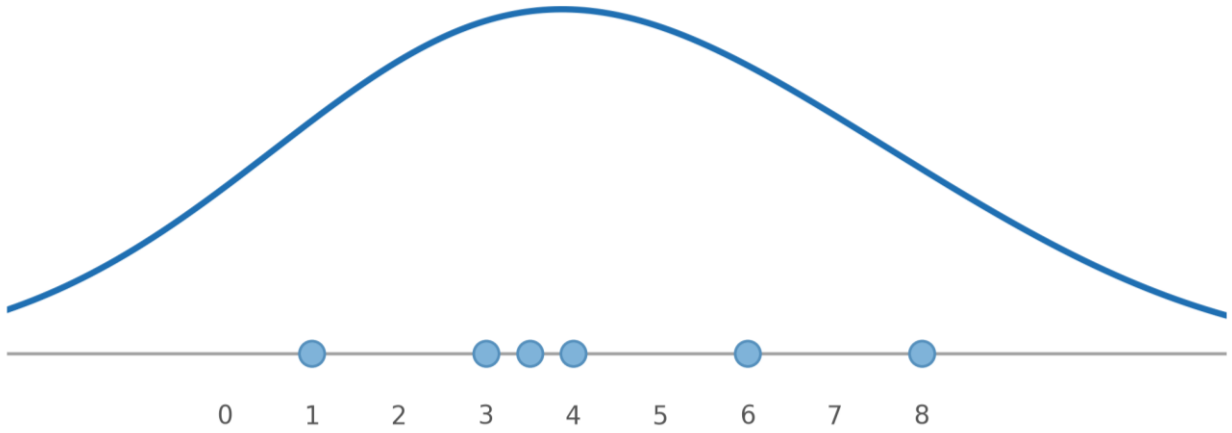


Density Estimation

How it works?

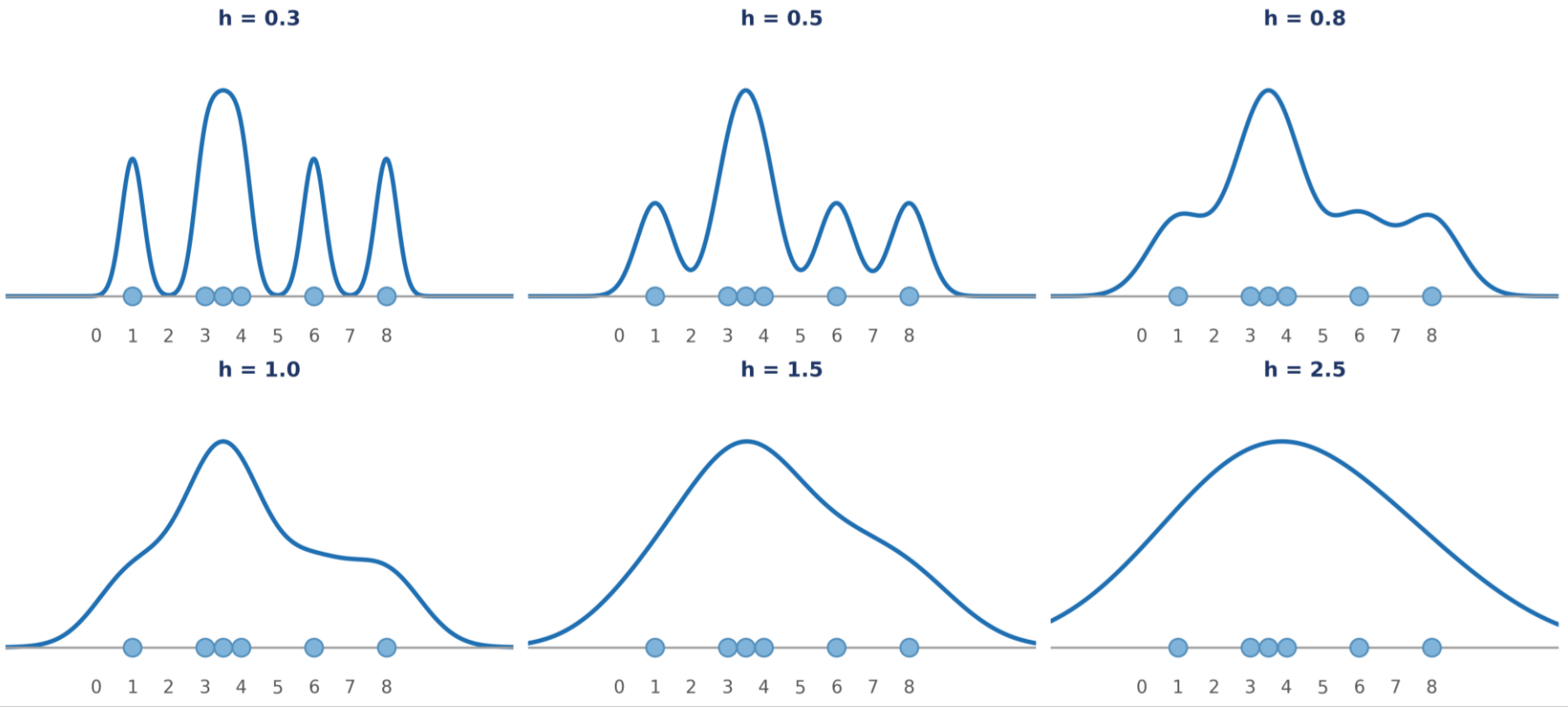
$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

h=2.5



Density Estimation

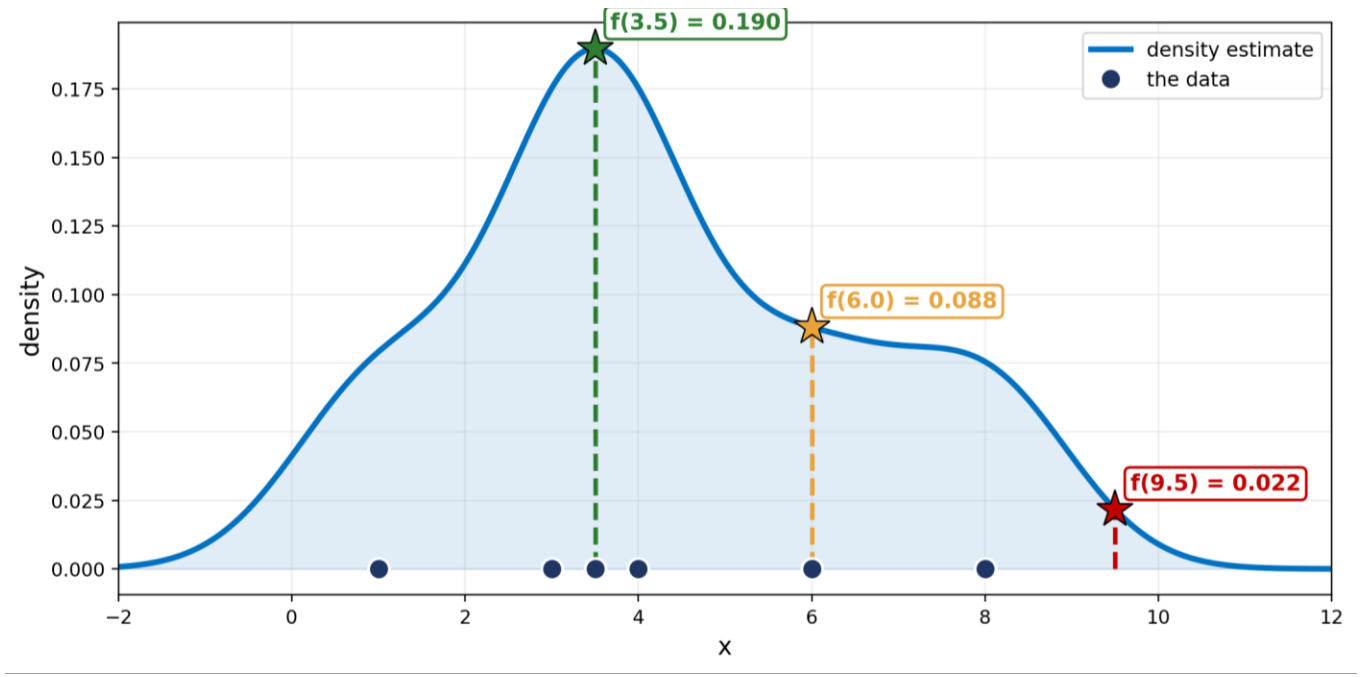
How it works?



Density Estimation

How it works?

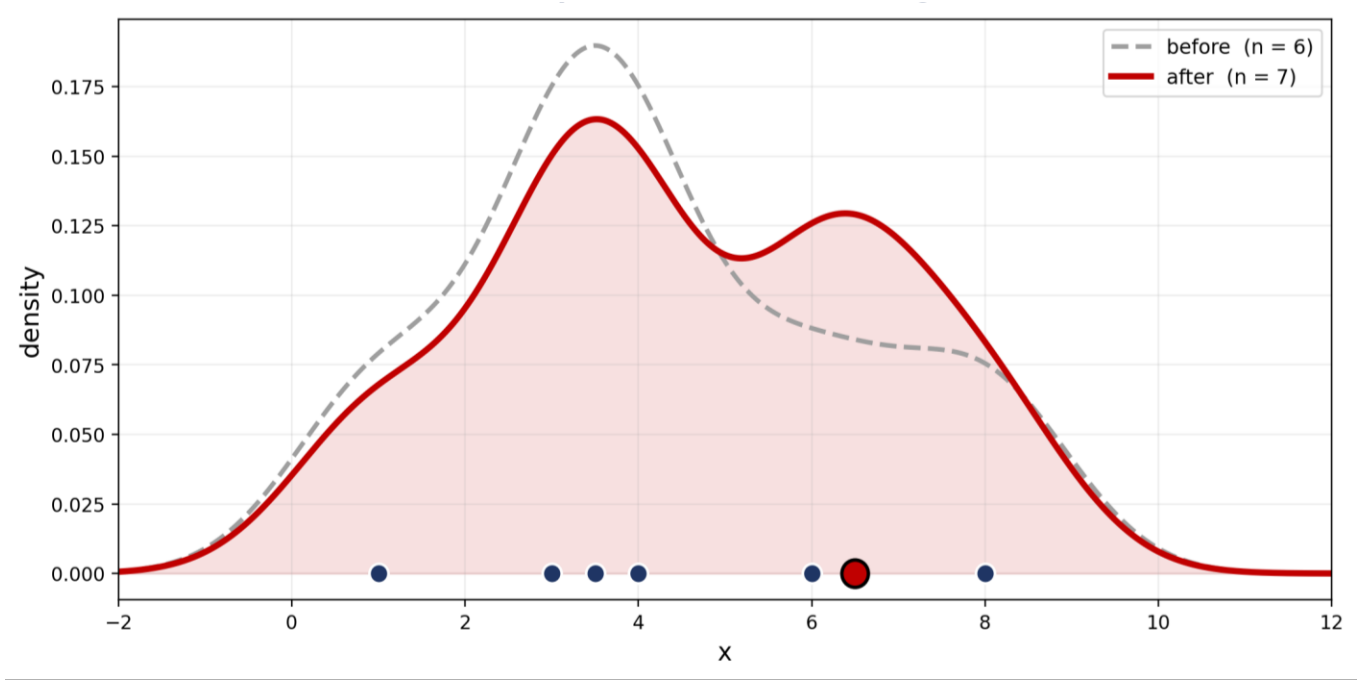
- **f(3.5) = 0.190.** The peak. Three of six points sit at 3, 3.5 and 4. The most crowded spot on the line.
- **f(6.0) = 0.088.** Less than half the peak. One point nearby, not three.
- **f(9.5) = 0.022.** Nearly nine times lower than the peak. Almost nothing within reach.



Density Estimation

How it works?

How density distribution changes with new datapoints are added .



DBSCAN

Definition

DBSCAN is a density-based method that finds clusters as dense regions of points separated by sparse regions, and labels points in the sparse regions as noise.

Chapter 7
Introduction to Data Mining, 2nd Edition

Where to use ?

- Fraud and anomaly detection. The noise pile is the interesting part.
- Geographic data. Finding real neighborhoods from GPS points.
- Any data where the groups are long, curved or irregular.

Pros?

- Finds the number of clusters by itself. You never choose k.
- Handles any shape. Long, curved, and irregular clusters are fine.
- Identifies noise explicitly. A point can belong to no cluster at all.

Cons?

- Slow. Comparing every pair does not scale past a few thousand rows.
- Once two clusters merge, they never separate. An early mistake is permanent.
- The linkage choice changes the answer a lot, and there is no correct one.

DBSCAN

How DBSCAN works?

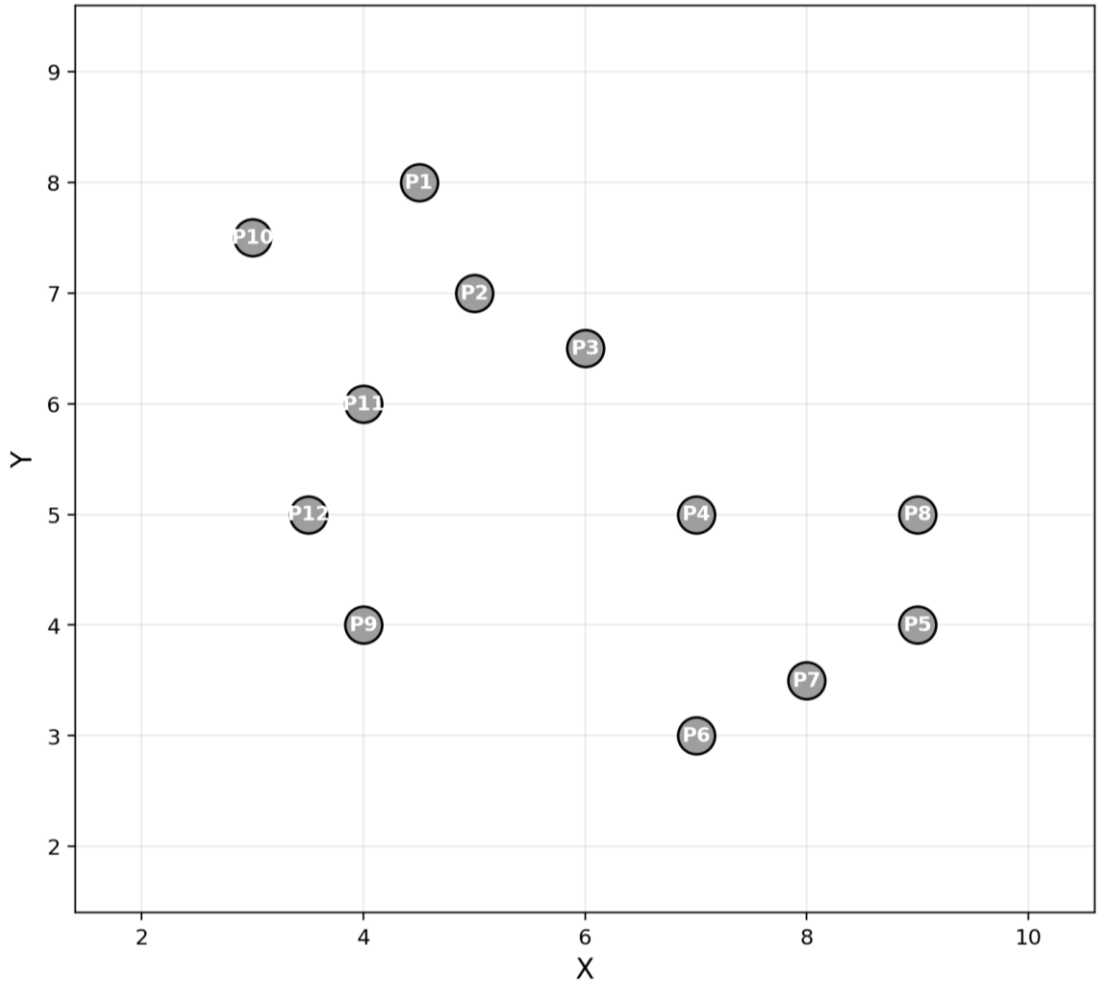
Point	X	Y
P1	4.5	8
P2	5	7
P3	6	6.5
P4	7	5
P5	9	4
P6	7	3
P7	8	3.5
P8	9	5
P9	4	4
P10	3	7.5
P11	4	6
P12	3.5	5

- **Step 1.** Compute the distance between every pair of points.
- **Step 2.** Choose two parameters: **eps**, the radius of the circle, and **minPts**, how many neighbors a point needs.
- **Step 3.** Draw a circle of radius eps around every point and count how many points fall inside, including itself.
- **Step 4.** Label every point:
  - **Core** if the count is at least minPts
  - **Border** if the count is below minPts, but the point sits inside some core point's circle
  - **Noise** if neither
- **Step 5.** Pick any unvisited core point and start a new cluster with it.
- **Step 6.** Add everything inside that core point's circle to the cluster.
- **Step 7.** If any newly added point is also a core point, add everything in its circle too. Keep going until nothing new joins.
- **Step 8.** Repeat from Step 5 with the next unvisited core point, until every core point is used.
- **Step 9.** Whatever is left over is noise. It belongs to no cluster.

DBSCAN

How DBSCAN works?

Point	X	Y
P1	4.5	8
P2	5	7
P3	6	6.5
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DBSCAN

Step 1. Compute the distance between every pair of points.

How DBSCAN works?

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P8	9	5
P9	4	4
P10	3	7.5
P11	4	6
P12	3.5	5

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$d=\sqrt{(6-5)^2+(6.5-7)^2}$$

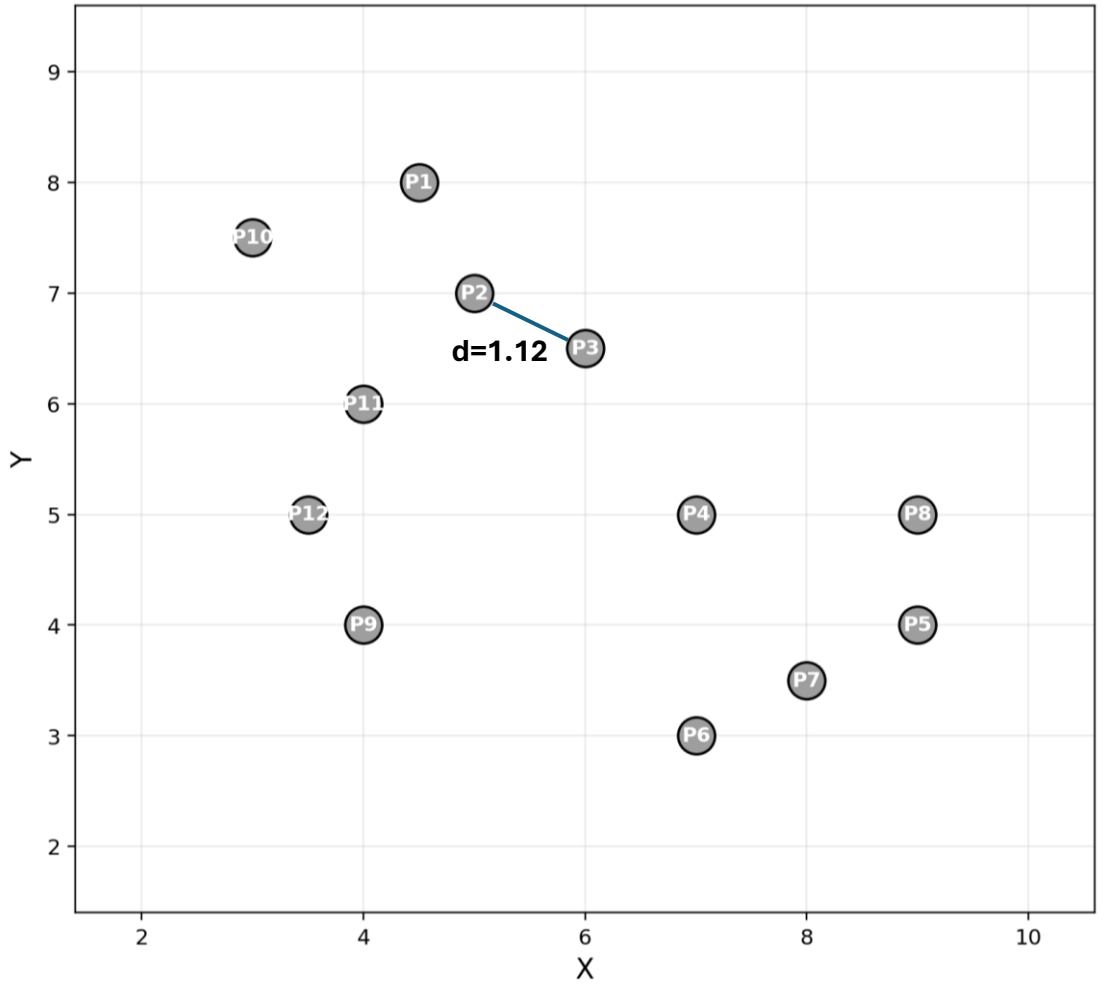
$$d=\sqrt{(1)^2+(-0.5)^2}$$

$$d=\sqrt{1+0.25}$$

$$d=\sqrt{1.25}$$

$$d=1.118$$

$$d=1.12$$



DBSCAN

How DBSCAN works?

Point	X	Y
P1	4.5	8
P2	5	7
P3	6	6.5
P4	7	5
P5	9	4
P6	7	3
P7	8	3.5
P8	9	5
P9	4	4
P10	3	7.5
P11	4	6
P12	3.5	5

Step 1. Compute the distance between every pair of points.

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	–	1.12	2.12	3.91	6.02	5.59	5.70	5.41	4.03	1.58	2.06	3.16
P2	1.12	–	1.12	2.83	5.00	4.47	4.61	4.47	3.16	2.06	1.41	2.50
P3	2.12	1.12	–	1.80	3.91	3.64	3.61	3.35	3.20	3.16	2.06	2.92
P4	3.91	2.83	1.80	–	2.24	2.00	1.80	2.00	3.16	4.72	3.16	3.50
P5	6.02	5.00	3.91	2.24	–	2.24	1.12	1.00	5.00	6.95	5.39	5.59
P6	5.59	4.47	3.64	2.00	2.24	–	1.12	2.83	3.16	6.02	4.24	4.03
P7	5.70	4.61	3.61	1.80	1.12	1.12	–	1.80	4.03	6.40	4.72	4.74
P8	5.41	4.47	3.35	2.00	1.00	2.83	1.80	–	5.10	6.50	5.10	5.50
P9	4.03	3.16	3.20	3.16	5.00	3.16	4.03	5.10	–	3.64	2.00	1.12
P10	1.58	2.06	3.16	4.72	6.95	6.02	6.40	6.50	3.64	–	1.80	2.55
P11	2.06	1.41	2.06	3.16	5.39	4.24	4.72	5.10	2.00	1.80	–	1.12
P12	3.16	2.50	2.92	3.50	5.59	4.03	4.74	5.50	1.12	2.55	1.12	–

DBSCAN

How DBSCAN works?

Point	X	Y
P1	4.5	8
P2	5	7
P3	6	6.5
P4	7	5
P5	9	4
P6	7	3
P7	8	3.5
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P9	4	4
P10	3	7.5
P11	4	6
P12	3.5	5

Step 2. Choose two parameters: **eps**, the radius of the circle, and **minPts**, how many neighbors a point needs

Eps: 1.8
minPts: 3

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	–	1.12	2.12	3.91	6.02	5.59	5.70	5.41	4.03	1.58	2.06	3.16
P2	1.12	–	1.12	2.83	5.00	4.47	4.61	4.47	3.16	2.06	1.41	2.50
P3	2.12	1.12	–	1.80	3.91	3.64	3.61	3.35	3.20	3.16	2.06	2.92
P4	3.91	2.83	1.80	–	2.24	2.00	1.80	2.00	3.16	4.72	3.16	3.50
P5	6.02	5.00	3.91	2.24	–	2.24	1.12	1.00	5.00	6.95	5.39	5.59
P6	5.59	4.47	3.64	2.00	2.24	–	1.12	2.83	3.16	6.02	4.24	4.03
P7	5.70	4.61	3.61	1.80	1.12	1.12	–	1.80	4.03	6.40	4.72	4.74
P8	5.41	4.47	3.35	2.00	1.00	2.83	1.80	–	5.10	6.50	5.10	5.50
P9	4.03	3.16	3.20	3.16	5.00	3.16	4.03	5.10	–	3.64	2.00	1.12
P10	1.58	2.06	3.16	4.72	6.95	6.02	6.40	6.50	3.64	–	1.80	2.55
P11	2.06	1.41	2.06	3.16	5.39	4.24	4.72	5.10	2.00	1.80	–	1.12
P12	3.16	2.50	2.92	3.50	5.59	4.03	4.74	5.50	1.12	2.55	1.12	–

DBSCAN

How DBSCAN works?

Step 2. Choose two parameters: **eps**, the radius of the circle, and **minPts**, how many neighbors a point needs

Eps: 1.8
minPts: 3

Point	X	Y
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P2	5	7
P3	6	6.5
P4	7	5
P5	9	4
P6	7	3
P7	8	3.5
P8	9	5
P9	4	4
P10	3	7.5
P11	4	6
P12	3.5	5

Neighbors within 1.8

P1, P2, P10
P1, **P2**, P3, P11
P2, **P3**
P4
P5, P7, P8
P6, P7
P5, P6, **P7**
P5, **P8**
P9, P12
P1, **P10**
P2, **P11**, P12
P9, P11, **P12**

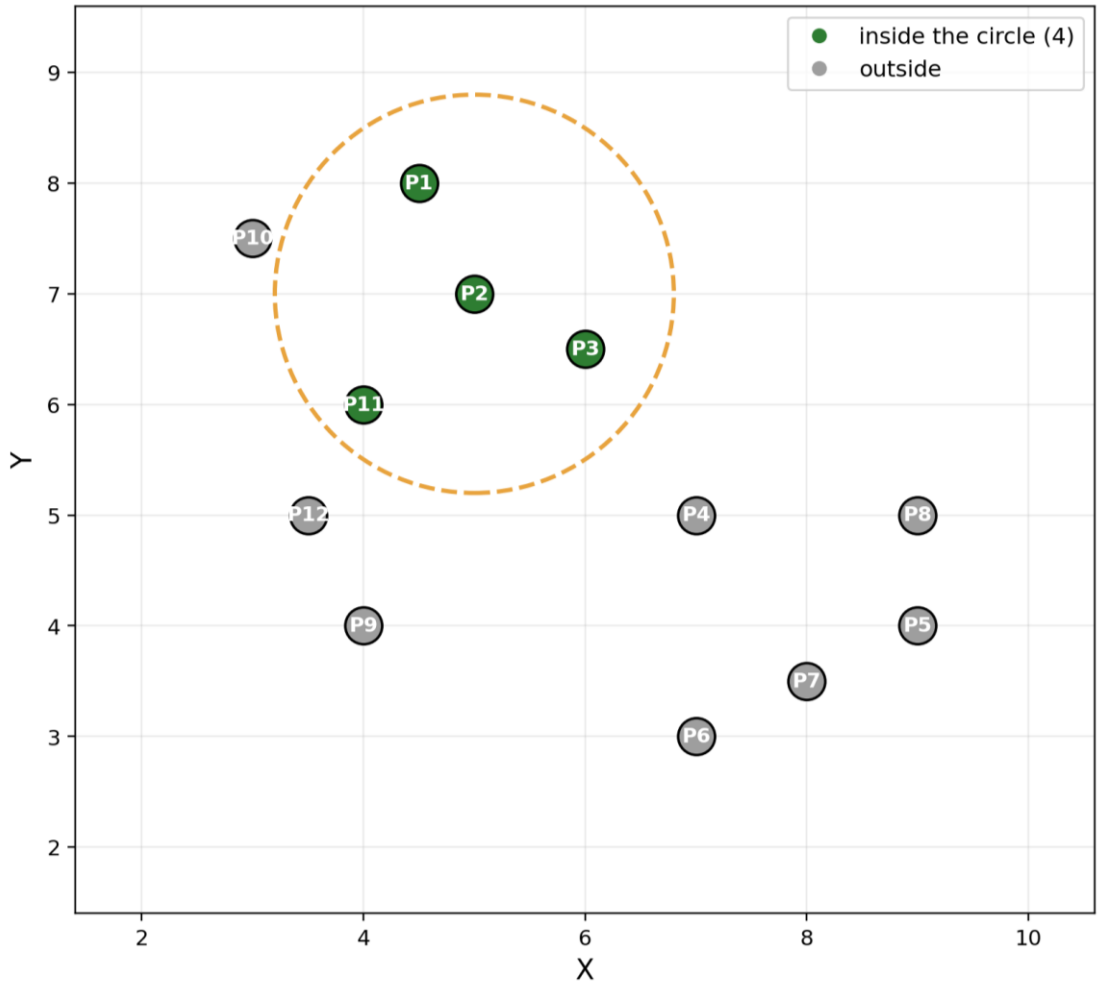
	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	–	1.12	2.12	3.91	6.02	5.59	5.70	5.41	4.03	1.58	2.06	3.16
P2	1.12	–	1.12	2.83	5.00	4.47	4.61	4.47	3.16	2.06	1.41	2.50
P3	2.12	1.12	–	1.80	3.91	3.64	3.61	3.35	3.20	3.16	2.06	2.92
P4	3.91	2.83	1.80	–	2.24	2.00	1.80	2.00	3.16	4.72	3.16	3.50
P5	6.02	5.00	3.91	2.24	–	2.24	1.12	1.00	5.00	6.95	5.39	5.59
P6	5.59	4.47	3.64	2.00	2.24	–	1.12	2.83	3.16	6.02	4.24	4.03
P7	5.70	4.61	3.61	1.80	1.12	1.12	–	1.80	4.03	6.40	4.72	4.74
P8	5.41	4.47	3.35	2.00	1.00	2.83	1.80	–	5.10	6.50	5.10	5.50
P9	4.03	3.16	3.20	3.16	5.00	3.16	4.03	5.10	–	3.64	2.00	1.12
P10	1.58	2.06	3.16	4.72	6.95	6.02	6.40	6.50	3.64	–	1.80	2.55
P11	2.06	1.41	2.06	3.16	5.39	4.24	4.72	5.10	2.00	1.80	–	1.12
P12	3.16	2.50	2.92	3.50	5.59	4.03	4.74	5.50	1.12	2.55	1.12	–

DBSCAN

How DBSCAN works?

Point	X	Y
P1	4.5	8
P2	5	7
P3	6	6.5
P4	7	5
P5	9	4
P6	7	3
P7	8	3.5
P8	9	5
P9	4	4
P10	3	7.5
P11	4	6
P12	3.5	5

Step 3. Draw a circle of radius eps around every point and count how many points fall inside, including itself.

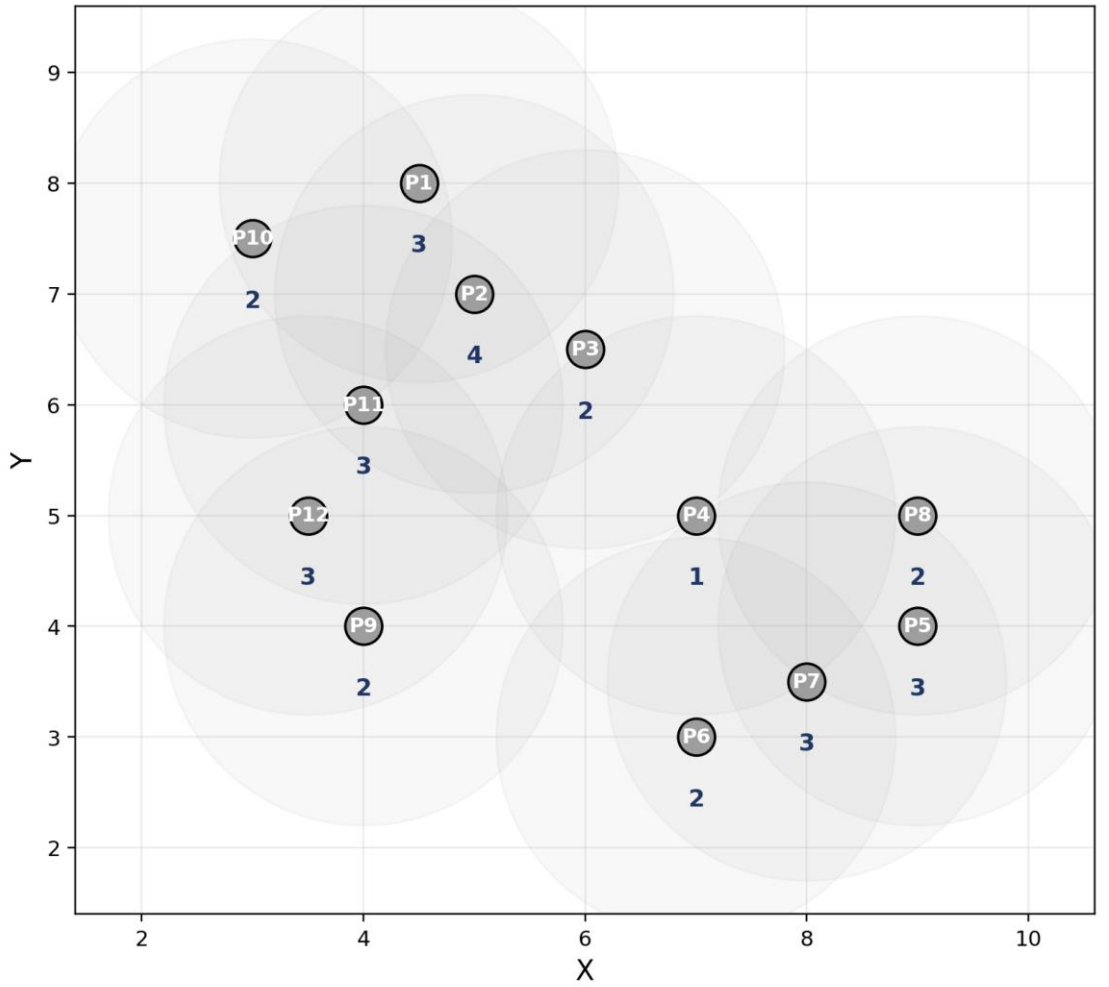


DBSCAN

How DBSCAN works?

Point	X	Y
P1	4.5	8
P2	5	7
P3	6	6.5
P4	7	5
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P7	8	3.5
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DBSCAN

How DBSCAN works?

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P2	5	7
P3	6	6.5
P4	7	5
P5	9	4
P6	7	3
P7	8	3.5
P8	9	5
P9	4	4
P10	3	7.5
P11	4	6
P12	3.5	5

Step 4. Label every point:

- **Core** if the count is at least minPts
- **Border** if the count is below minPts, but the point sits inside some core point's circle
- **Noise** if neither

Point	Neighbours within 1.8	Count	Type
P1	P1, P2, P10	3	**core**
P2	P1, P2, P3, P11	4	**core**
P3	P2, P3	2	border
P4	**P4**	**1**	**NOISE**
P5	P5, P7, P8	3	**core**
P6	P6, P7	2	border
P7	P5, P6, P7	3	**core**
P8	P5, P8	2	border
P9	P9, P12	2	border
P10	P1, P10	2	border
P11	P2, P11, P12	3	**core**
P12	P9, P11, P12	3	**core**

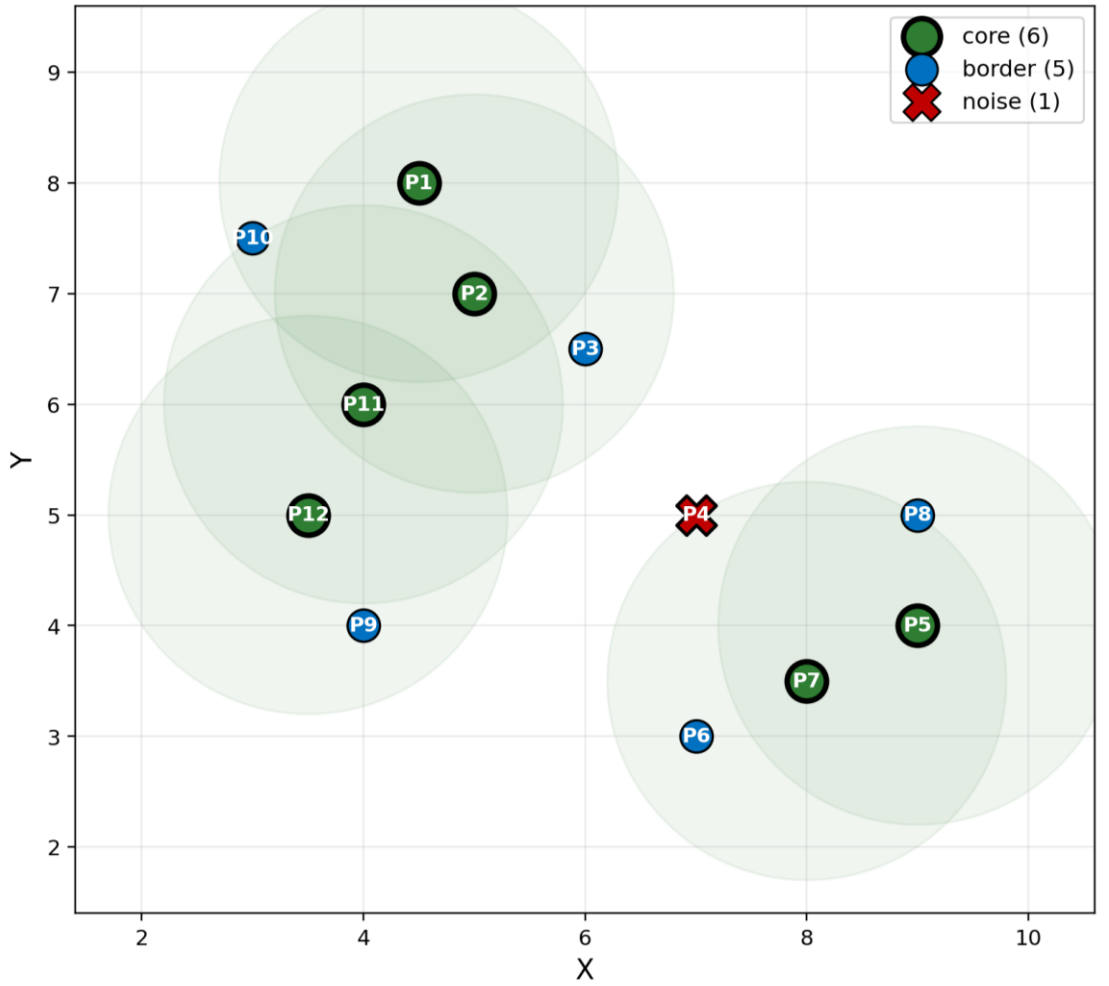
DBSCAN

How DBSCAN works?

Point	X	Y
P1	4.5	8
P2	5	7
P3	6	6.5
P4	7	5
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P8	9	5
P9	4	4
P10	3	7.5
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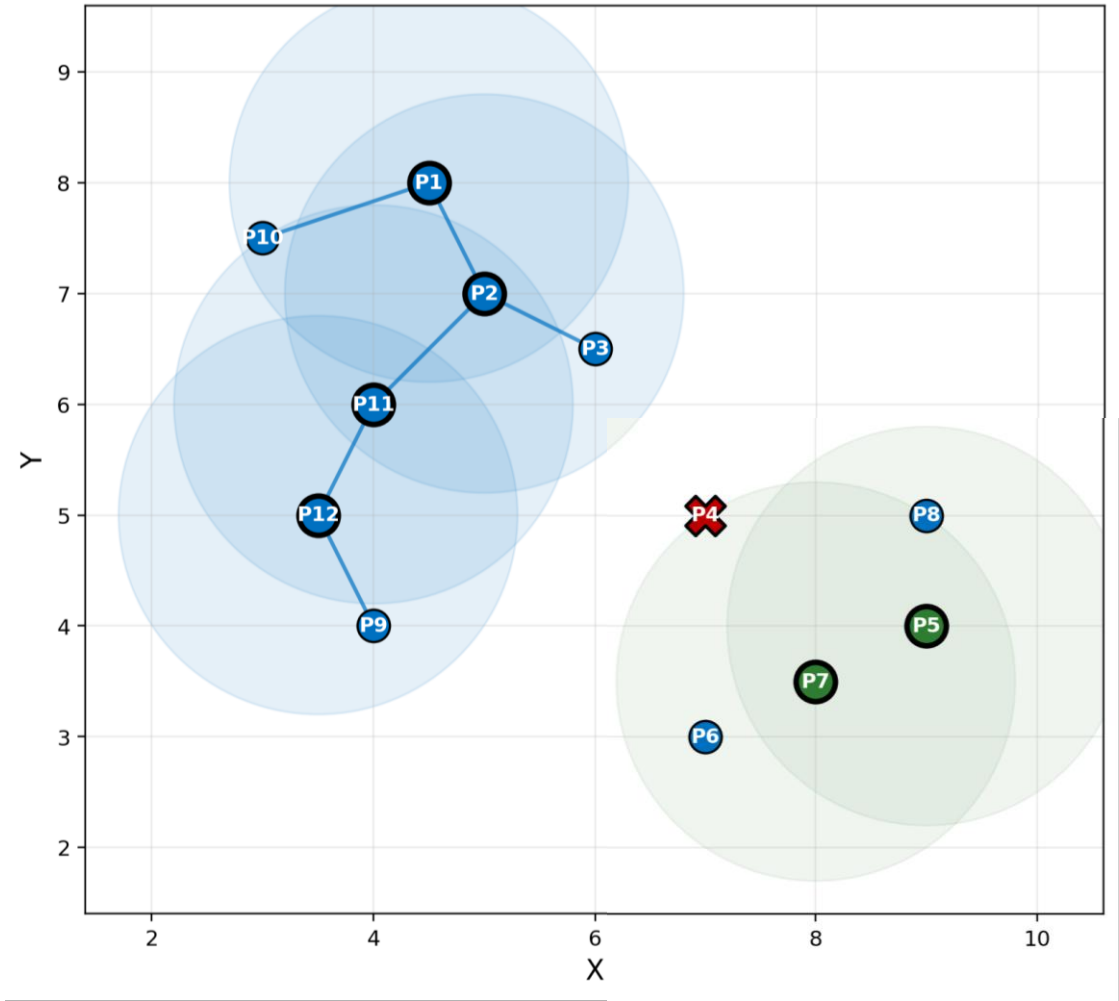


DBSCAN

How DBSCAN works?

Point	X	Y
P1	4.5	8
P2	5	7
P3	6	6.5
P4	7	5
P5	9	4
P6	7	3
P7	8	3.5
P8	9	5
P9	4	4
P10	3	7.5
P11	4	6
P12	3.5	5

Step 5. Pick any unvisited core point and start a new cluster with it.
Step 6. Add everything inside that core point's circle to the cluster.
Step 7. If any newly added point is also a core point, add everything in its circle too. Keep going until nothing new joins.

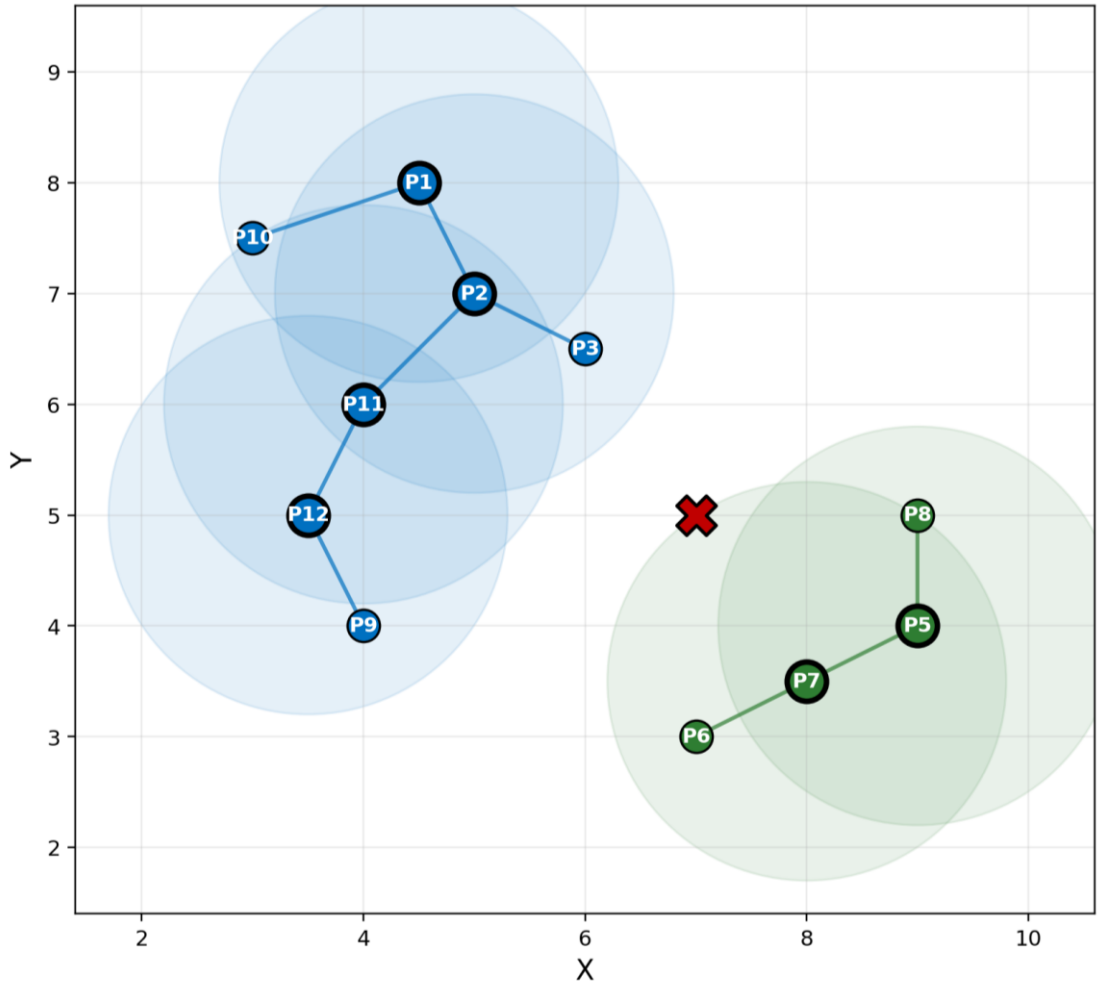


DBSCAN

How DBSCAN works?

Point	X	Y
P1	4.5	8
P2	5	7
P3	6	6.5
P4	7	5
P5	9	4
P6	7	3
P7	8	3.5
P8	9	5
P9	4	4
P10	3	7.5
P11	4	6
P12	3.5	5

Step 8. Repeat from Step 5 with the next unvisited core point, until every core point is used.

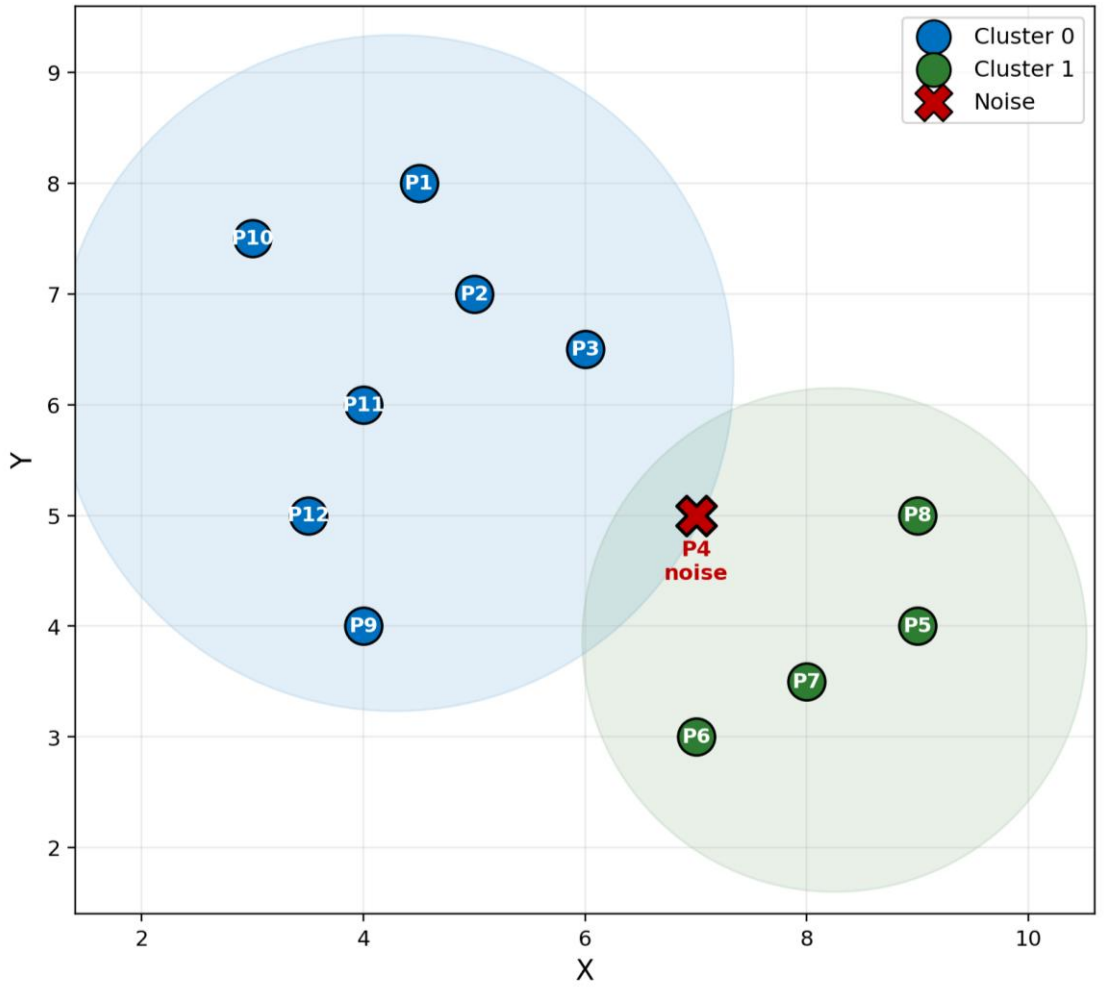


DBSCAN

How DBSCAN works?

Point	X	Y
P1	4.5	8
P2	5	7
P3	6	6.5
P4	7	5
P5	9	4
P6	7	3
P7	8	3.5
P8	9	5
P9	4	4
P10	3	7.5
P11	4	6
P12	3.5	5

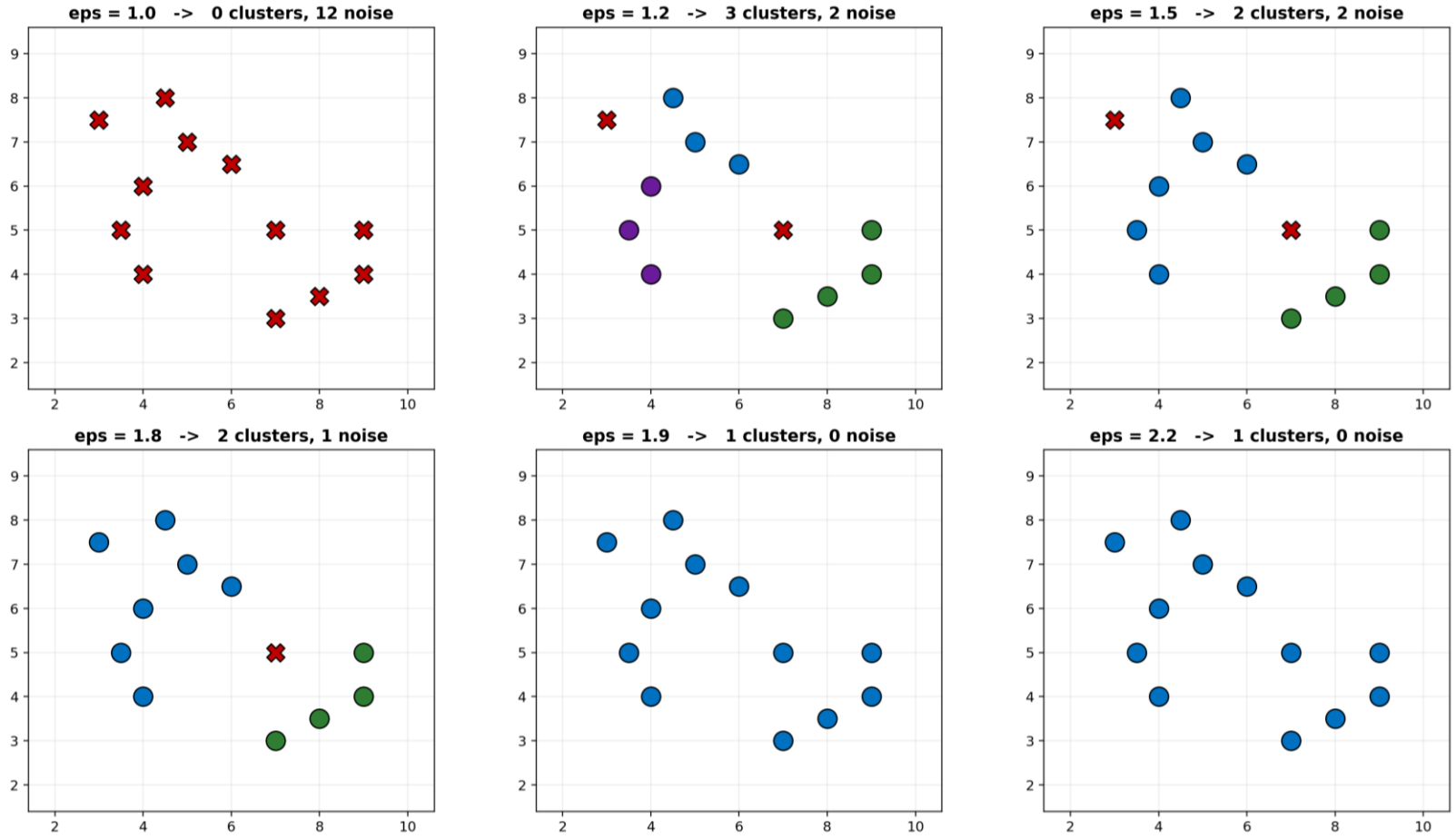
Step 9. Whatever is left over is noise. It belongs to no cluster.



DBSCAN

How DBSCAN works?

Point	X	Y
P1	4.5	8
P2	5	7
P3	6	6.5
P4	7	5
P5	9	4
P6	7	3
P7	8	3.5
P8	9	5
P9	4	4
P10	3	7.5
P11	4	6
P12	3.5	5



DBSCAN on Ames

Our Question, and What the Code Will Do

- **The question**

  “Forget how many groups there are. Which houses sit in dense neighbourhoods, and which ones are so unusual they belong to no group at all?”

- **What the code will do**

  Scale first, then try several values of eps.
  Count how many clusters DBSCAN finds on its own.
  Look at the houses it labels as noise. Are they really unusual?
  Compare the three methods on the same houses.

- **This is the only method today that is allowed to answer “none of the above.”**